

pandas!!!



Two Months Salary





What's so special about diamonds?



What's so special about diamonds?



Each one is said to be unique.

The Four C's:

- Carat
- Color
- Clarity
- Cut



What's so special about diamonds?

The perfect gift to represent engagement.

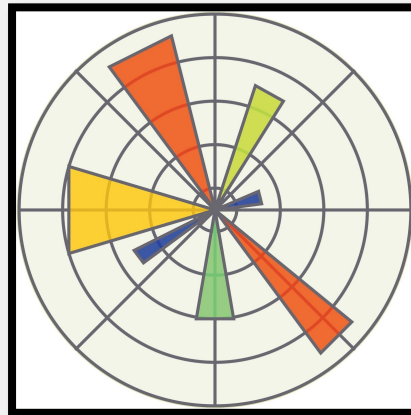
Traditionally should cost **two months salary**.



Data Analysis in Python



pandas and matplotlib



Data Analysis in Python



Rarely conducted from scratch.

Generally faster and easier if done using **open source** packages.

What is open source?



<https://opensource.org/>

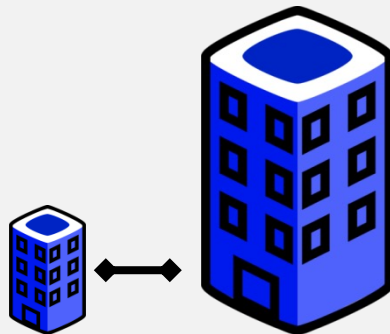
The Great Divide



Pre-Business Intelligence

Data was disorganized and rarely taken advantage of.

Lack of computer science technology was a **major** barrier.



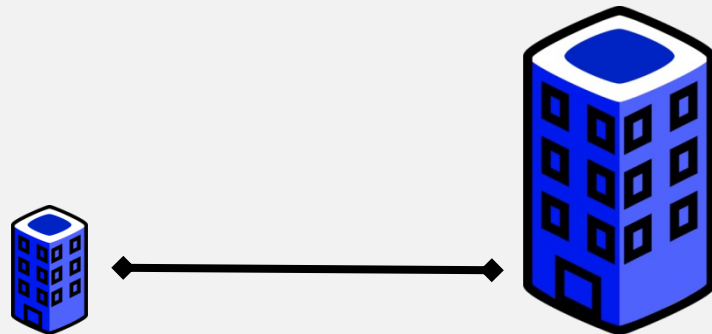
The Great Divide



Business Intelligence

Large companies could afford expensive technological solutions.

Data became organized and adopters had a better picture as to what was going on.



The Great Divide

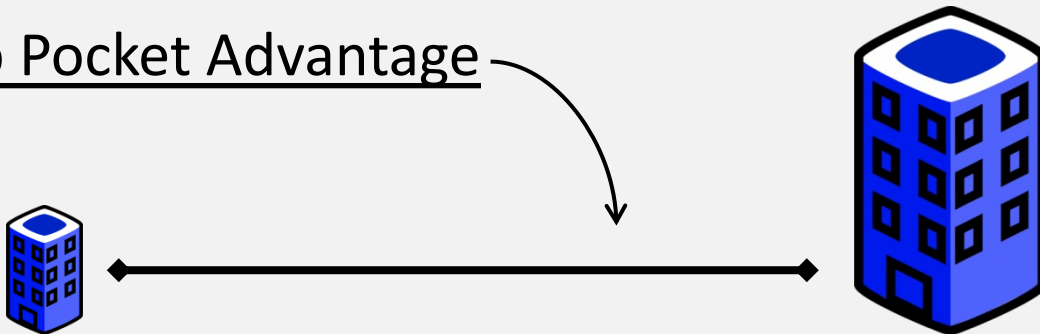


Analytics

Data was disorganized and rarely taken advantage of.

Lack of computer science technology was a **major** barrier

The Deep Pocket Advantage



The Death of the Deep Pocket Advantage



Enterprise v. Open Source



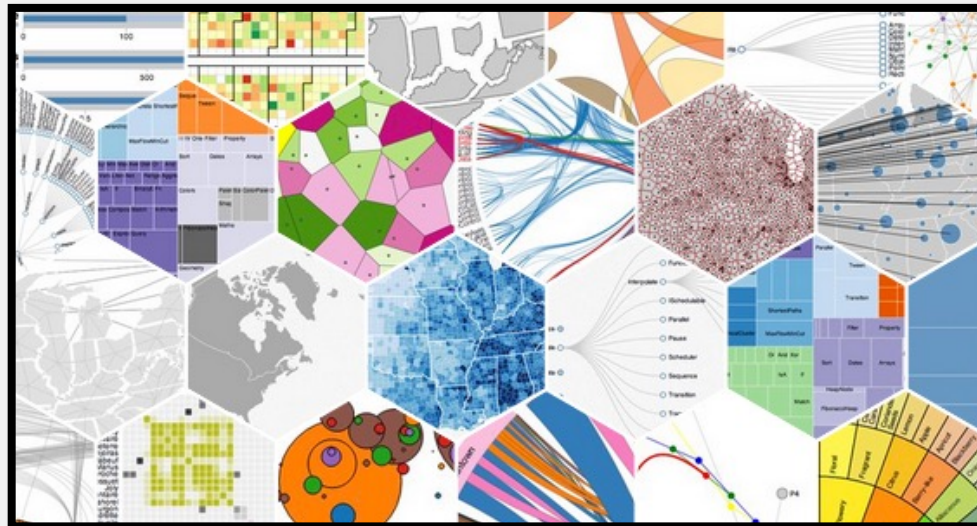
Enterprise
Tableau

Data
Visualization

Open Source

D3

(Data-Driven Documents)

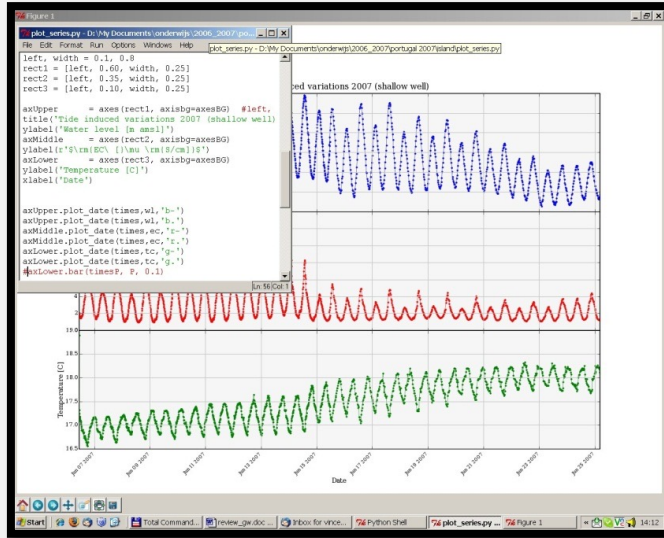


Enterprise v. Open Source

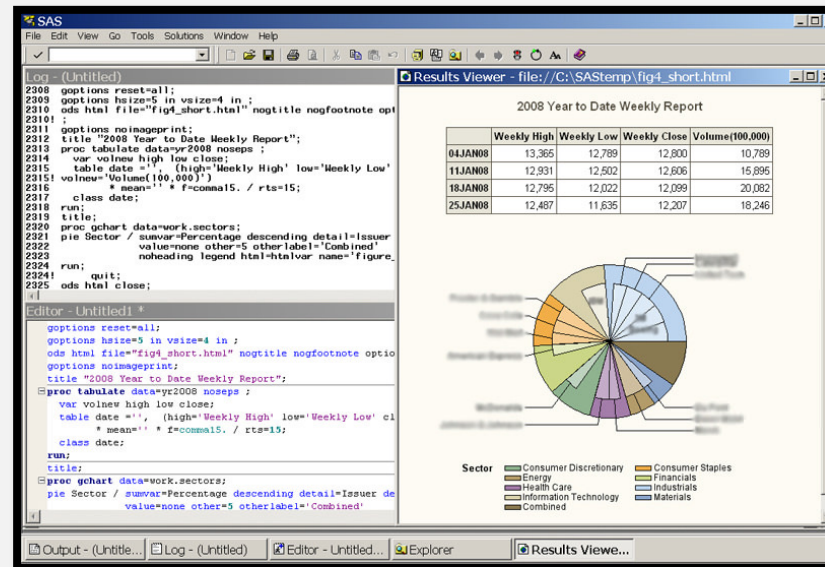


Open Source
Python

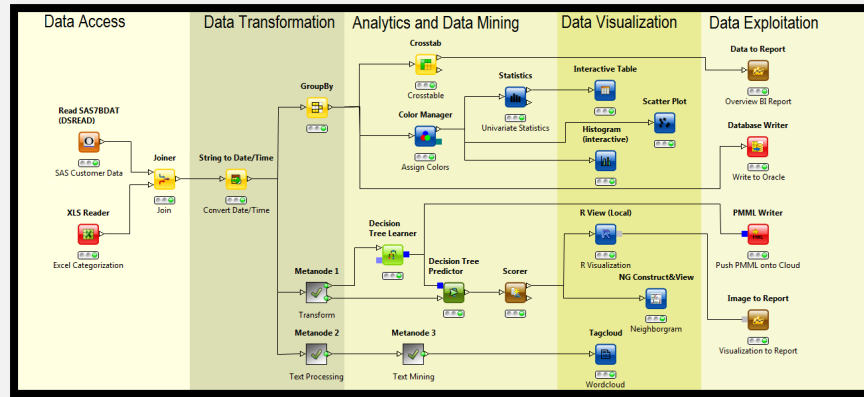
Computational
Programming



Enterprise
SAS



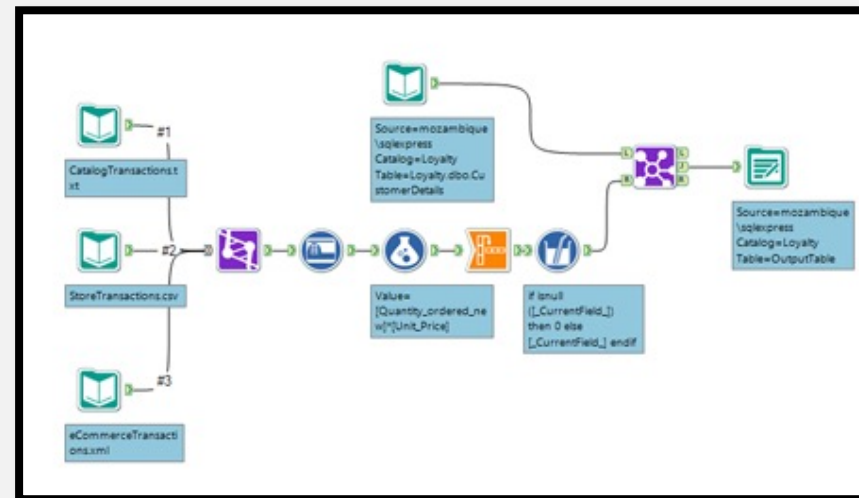
Enterprise v. Open Source



**Open Source
Kettle**

**Data Cleansing
and Mining**

**Enterprise
Alteryx**



pandas



Data Science Essentials



What is pandas?



World class open source project

Addresses Python's need for data analysis

Home to the **DataFrame**, Python's version of a **spreadsheet**

pandas



Generally imported as **pd**

```
In [ ]: 1 import pandas as pd
```

This makes it easier to reference pandas functions and sub-packages

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Jupyter Notebook will list available functions in ***most*** packages.

```
In [ ]: 1 pd.
```

- api
- array
- arrays
- bdate_range
- Categorical
- CategoricalDtype
- CategoricalIndex
- compat
- concat
- core



pandas Practice

Accessing Packages and Options



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How many **mandatory** arguments does **pd.to_numeric()** have?

```
In [4]: 1 help(pd.to_numeric)
```

```
Help on function to_numeric in module pandas.core.tools.numeric:
```

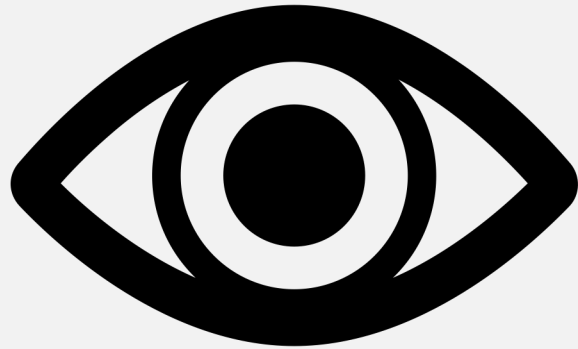
```
to_numeric(arg, errors='raise', downcast=None)  
    Convert argument to a numeric type.
```

Just one.



Viewing our Dataset

Understanding What's There



Useful Techniques (1 of 2)



df (pd.DataFrame) is a generic way to say dataset

Command	Description
df	accesses several rows and columns of a dataset
df.head()	accesses the first <i>n</i> rows in a dataset
df.tail()	accesses the last <i>n</i> rows in a dataset
df.shape()	accesses the dimensions of a dataset
df.columns	accesses the column names in a dataset
df.count()	counts the number of non-missing values in a dataset

Useful Techniques (2 of 2)



df (pd.DataFrame) is a generic way to say dataset

Command	Description
<code>df[df[col] > x]</code>	accesses rows of a dataset that meet specified criteria
<code>df. .sort_values([cols], ascending=True)</code>	sorts the values of a given dataset
<code>df.info()</code>	accesses column names, number of non-missing values, and variable types
<code>df.describe()</code>	accesses the mean and quantiles of a dataset
<code>df.loc[rows, cols]</code>	accesses specified rows and columns of a dataset (numpy required)
<code>df.iloc[row #'s, col #'s]</code>	accesses specified rows and columns of a dataset (numpy required)

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